

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

RI2105-1 – DATA STRUCTURE AND ALGORITHMS

(RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

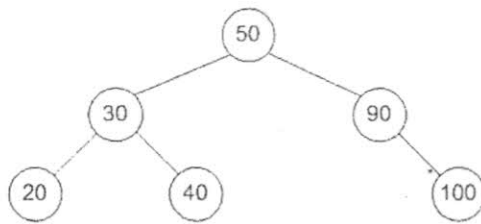
- Which of the following is not the type of queue?
 A) Priority queue B) Circular queue
 C) Single ended queue D) Ordinary queue
- Which data structure is used for implementing recursion?
 A) Stack B) Queue
 C) List D) Array
- What is the worst case time complexity of inserting a node in a doubly linked list?
 A) $O(n \log n)$ B) $O(\log n)$
 C) $O(n)$ D) $O(1)$
- What is the data type of the array passed from the command line into the main() function in C?
 A) char arr[]; B) char *arr[];
 C) char **arr[]; D) char arr[][];
- Pointer cannot be used to
 A) Find the address of a variable in memory B) Reference value directly
 C) Simulate call by reference D) Manipulate dynamic data structure
- What is the purpose of -> operator in C when used with pointers?
 A) Member access operator B) Indirection operator
 C) Bitwise operator D) Arithmetic operator
- What is the need for a circular queue?
 A) easier computations B) implement LIFO principle in queues
 C) effective usage of memory D) to delete elements based on priority
- Which of the following is a non linear data structure
 A) Stack B) Queue
 C) Trees D) Arrays
- The number of edges from the root to the node is called _____ of the tree.
 A) Height B) Depth
 C) Width D) Length
- What is a full binary tree?
 A) Each node has exactly zero or two children B) Each node has exactly two children
 C) All the leaves are at the same level D) Each node has exactly one or two children
- Which of the following properties does a simple graph not hold?
 A) Must be connected B) Must be unweighted
 C) Must have no loops or multiple edges D) Must have no multiple edges
- A graph with all vertices having equal degree is known as a _____.
 A) Multi Graph B) Regular Graph
 C) Multi Graph D) Complete Graph

13. Which of the following asymptotic notations is used to provide lower bound constraint?
 A) x B) Θ
 C) Ω D) O
14. What is recursion?
 A) Calling the main function B) Calling inbuilt function
 C) Calling other function D) Calling the same function again and again within itself
15. What is the worst case time complexity if element not found in linear search
 A) $O(1)$ B) $O(n)$
 C) $O(-1)$ D) $O(n-1)$
16. Base condition in recursive algorithm is also called as
 A) Starting condition B) Intermediate condition
 C) terminating condition D) Stopping condition
17. Algorithm in which concept of backtracking is used
 A) N Queens problem B) Dijkstra's algorithm
 C) Kruskal's algorithm D) Prism Algorithm
18. Which of the following algorithm design techniques is used in the quick sort algorithm
 A) Decrease and conquer B) Dynamic programming
 C) Greedy Method D) Divide and conquer
19. Which of the following is not a stable sorting algorithm?
 A) Insertion sort B) Selection sort
 C) Bubble sort D) Merge sort
20. What is the worst case complexity of bubble sort?
 A) $O(n \log n)$ B) $O(\log n)$
 C) $O(n)$ D) $O(n^2)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the differences between primitive and non-primitive data types with a neat diagram and explain various operations of data structures.	8	L*2	1	2
	b) Explain in detail concept of singly linked list. Develop a C program to demonstrate implementation of singly linked list.	8	L3	2	2
2.	a) Define pointers explain pictorially concept of pointers and write a program to explain concept of pointers with functions.	8	L3	1	2
	b) Define queue. explain its operations. Write a C program to implement queue and its operations.	8	L3	2	2
3.	a) Explain array of structures with neat diagram, write a C program the demonstrates relationship between structures and pointers.	8	L2	1	2
	b) Write an algorithm for conversion of infix to postfix expression. Demonstrate for the following example. A+(B*C-(D/E-F)*G)*H	8	L2	2	2
Unit – II					
4.	a) With a suitable example explain insertion, simple deletion, and traversals in binary trees with a C program.	8	L3	3	2
	b) Give the two conditions for successful recursive function. Design a recursive function to solve tower o Hanoi problem and show tracing it for number of disk = 3.	8	L2	4	2

5. a) Explain binary tree and its key properties, demonstrate in order, preorder, post order tree traversals along with their C functions for the given tree in FigQ5(a)



FigQ5(a)

- b) Write a quick sort algorithm to sort the following elements and illustrate its time complexity for the given numbers. 7, -2, 4, 1, 6, 5, 0, -4, 2.
6. a) Explain the linear and linked representations of a binary tree, various ways of graph representations.
- b) Explain with neat diagram in detail three asymptotic notations with example for each, and its significance in analyzing algorithms.

8 L3 3 2

8 L3 4 2

8 L2 3 2

8 L2 4 3

Unit – III

7. a) Explain prims algorithm with the suitable example.
- b) Explain decrease and conquer techniques with the suitable example.
8. a) Explain N-Queens problem with the suitable example.
- b) Explain greedy method with the suitable example and its respective algorithm.

8 L2 5 3

8 L2 5 3

8 L2 5 3

8 L2 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
